Indian Institute of Engineering Science and Technology, Shibpur
BTech 4^{th} Semester Semester (CST) Examination, May, 2026
Theory of Computation (CS2204)

Full Marks: 50 **Time: 3 Hour**

Answer Question No - 1 and any Four from the remaining.

1. (a) Prove that reversal of any regular language is also regular.
   (b) State Pumping Lemma for Context Free Languages.
   (c) What do you mean by useless symbol? Explain with an example.
   (d) Find the language generated by the following grammar productions: $S \rightarrow AB, A \rightarrow A1|0, B \rightarrow 2B|3$. here S is the start symbol. Can this language be generated by a grammar of higher type?
   (e) Construct a finite automaton for the regular expression $(a+b)^*abb$.

[$2+2+2+2+2=10$]

2. (a) Prove the following identity:

$$(a^*ab+ba)^*a^* = (a+ab+ba)^*$$

   (b) Construct a finite automaton recognizing L(G), where G is the grammar with the following set of production rules:

$$S \rightarrow aS|bA|b, A \rightarrow aA|bS|a$$

   (c) Show that there exists no finite automaton accepting all palindromes over $\{a,b\}$.

[$3+3+4=10$]

3. (a) Consider the following productions where S is the start symbol:

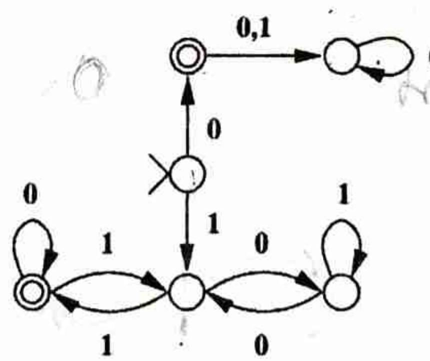
$$S \rightarrow aB|bA, A \rightarrow aS|bAA|a, B \rightarrow bS|aBB|b$$

   Here, {S, A, B } are variables. For the string $aaabbabbba$, find
   i. the leftmost derivation and
   ii. the rightmost derivation,
   iii. the parse tree and test whether the grammar is ambiguous or not.

   (b) Given the grammar with following production rules: $S \rightarrow AB, A \rightarrow a, B \rightarrow C|b, C \rightarrow D, D \rightarrow E, E \rightarrow a$, find an equivalent grammar which is reduced and has no unit productions.

[$(2+2+2)+4=10$]

4. (a) Give an informal description of the language accepted by the following DFA.



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   (b) Let $\Sigma = \{a,b\}$ and $L = \{\alpha \in \Sigma^* | \alpha$ ends with bb and but does not start with bb$\}$
   i. Write a regular expression (over Σ) to represent L.
   ii. Design a DFA whose language is L.

[$4+(2+4)=10$]

5. (a) State Pumping Lemma for CFL. Use Pumping lemma to show that the following language is not a CFL: $L = \{a^n b^n c^n | n \geq 1\}$.

(b) Construct Pushdown Automata that accept the following the language over the alphabet $\{a, b, c\}$: $L = \{wcw^R | w \in \{a, b\}^*$where c is the middle symbol$\}$. Show an instantaneous description for the following input string $abcba$.

[(2 + 3) + 5 = 10]

6. (a) Reduce the following grammar to Chomsky Normal Form:

$$S \to AB, A \to BSB, A \to BB, B \to aAb, B \to a, A \to b$$

Here, S is the start symbol, uppercase letters are variables, and lowercase letters are set of terminals.

(b) Draw a Turing Machine that accepts the language $L = \{x \in \{a, b\}^* | n_a(x) = n_b(x) = n_c(x)\}$. Show an instantaneous description for the following input string $aabbcc$.

S → A~~b~~aB